

with original polarisation technology are used to produce high-sensitivity and excellent durability. These are of re-flow solderable surface-mount device type and boast of high resonant frequency and wide-bandwidth operation.

Sample application idea. Write-protect your hard disk data, when it is under shock from the outside; enable pick-up control for disk-type storage in digital cameras and camcorders; detect minute vibrations of machinery; detect rapid acceleration (automotive); detect vibrations due to destructive acts (commercial windows, vending machines, etc)

Board-Mount Barometric Pressure Sensors from Murata

Murata ZPA series micro-electro-mechanical systems (MEMS) barometric pressure sensor is a capacitive-type sensor offering low current consumption and stable temperature drift.

ZPA series' capacitive MEMS technology is suitable for applications that require high accuracy and continuous data extraction. Compared to the existing piezo resistive type, low noise performance and low current consumption can be achieved simultaneously with the barometric pressure sensor.

Users can select output data rate from 1Hz, 5Hz, 11Hz to 23Hz. Change in pressure value will be reflected if the height is changed, the breath is blown in or the door is opened under closed environment conditions.

Sample application idea. Why not embed it in your mobile navigation device before your next trek?

UMAC Series Small Energy Device from Murata

A part of the UMAC series, the small energy device is a secondary battery, with features similar to that of a capacitor. Designed to be maintenance-free, it has a maximum charging/discharging rate of 10C (30mA) and an extended service cycle life with a

charging recovery rate of over 80 per cent, even after 1.5K cycles. It is a safe device as no thermal runaway occurs because of its low capacity and chemically-stable materials.

Sample application idea. Ideal for a small power equipment like an electric pen, a solar battery charger equipment and sensor node with wireless sensor network in combination with an energy-harvesting system

Type YD-Certified Wi-Fi Module from Murata

This Wi-Fi module is an embedded network controller solution providing simple serial-to-Wi-Fi Internet connectivity. It is type-YD-certified, which gives it the ability to IP-enable any device with a UART or SPI interface, integrating STM32 ARM Cortex-M3 microcontroller with additional interfaces.

The network controller module provides device manufacturers with an easy-to-design solution for data acquisition, device management and industrial control applications.

Sample application idea. Put together a sensor device application using ADC and GPIO interfaces and send data to the cloud via Wi-Fi connectivity

Bluetooth Smart Sensor Development Kit from Broadcom

Broadcom's WICED Sense Bluetooth smart sensor development kit is a kit for WICED SMART product family based on BCM20737S SIP module, featuring Bluetooth Smart technology. With six MEMs sensors on board, customers can quickly ramp up a demo/prototype specific to their needs. The over-the-air download feature provides the ability to update the tag's firmware from a central device like a smartphone, tablet or PC.

Sample application idea. With a development kit at your disposal that includes an associated smartphone app, why not develop a Bluetooth Smart app/demo

Bluetooth Smart Low Energy Module from Broadcom

Broadcom BCM20737S WICED Smart Bluetooth Low Energy module allows manufacturers to seamlessly connect peripheral devices like heart rate monitors, pedometers, door locks, lighting, proximity alarms and more. Designed for simple integration into coin-cell batteries, ARM Cortex M3 based BCM20737S will inspire a new range of connectivity for previously-unconnected devices.

Sample application idea. How about connecting to a headless device without a means of interactive configuration

Bluetooth Smart SoC by Broadcom

Broadcom's BCM20736 WICED SMART Bluetooth system on chip (SoC) provides a flexible solution that can be incorporated into a wider variety of devices to drive new use cases. The chip includes built-in wireless charging support for Alliance for Wireless Power (A4WP) standard.

Additionally, the highly-integrated design and smaller size of BCM20736 reduces power consumption to extend the battery life for wearables.

Sample application idea. Let us leave you to figure this one out. Maybe you could use a combination of modules

Enable your creativity today!

And, so we are done. Challenging enough? Get on to it then. You can work individually or in a team of up to four members. Registration is free. Top-100 projects get through to the next round.

All you need to do now is form your team, register, cook up your idea, send in your abstract and wait. For any other details, log on to <http://mouser.electronicsforu.com>

Do not forget, you just have till the 12th February 2016, to submit your abstract.